

1. Introduction

In my recent studies in the Tryhackme Python Basics room, I have learned the basics of Python programming, a versatile and user-friendly language. This foundational knowledge includes:

Variables: Storing and manipulating data.

Loops: Repeating tasks and iterating over data.

Functions: Writing reusable code blocks.

Data Structures: Organizing data with lists, dictionaries, tuples, and sets.

If Statements: Controlling program flow based on conditions.

Files: Reading from and writing to files for data management.

I have come to understand that these concepts are crucial for scripting and automation in cybersecurity thus forming a solid foundation for my future work in the field.

2. Module Sections

I typed `print("Hello World")` on the `script.py` terminal and got the flag.

The image displays two screenshots of the TryHackMe Python Basics room interface, showing the progression from an initial state to a completed exercise.

Top Screenshot (Initial State):

- Room progress (5%):** The progress bar is at 5%.
- Code Editor:** The code editor shows a single line of Python code: `print("Hello World")`.
- Python code output:** The output shows the result of running the code: `Learn security with TryHackMe!` and `Hello World`.
- Exercise Complete! The flag is:** `THM(PRINT_STATEMENTS)`

Bottom Screenshot (Completed State):

- Room progress (11%):** The progress bar is at 11%.
- Code Editor:** The code editor shows the same code: `print("Hello World")`.
- Python code output:** The output shows the result of running the code: `Learn security with TryHackMe!` and `Hello World`.
- Exercise Complete! The flag is:** `THM(PRINT_STATEMENTS)`
- Confirmation Message:** A green message box says: "Woop woop! Your answer is correct".
- Answer Input:** The input field contains the correct answer: `THM(PRINT_STATEMENTS)`.
- Buttons:** There are two buttons: "Correct Answer" (green) and "Hint" (orange).

I typed `print(21+43)` in the code editor and got the flag

The image displays two screenshots of the TryHackMe Python Basics room interface, showing the progression from a coding exercise to a successful submission.

Top Screenshot (Room progress: 11%):

- Left Panel (Questions):**

Not Equal to	<code>!=</code>
Greater than or equal to	<code>>=</code>
Less than or equal	<code><=</code>

Answer the questions below

In the code editor, print the result of 21 + 43. What is the flag?

Print the result of 142 - 52. What is the flag?
- Right Panel (Code Editor):**
 - Files: `script.py`, `flag.txt`, `shipping.py`, `bitcoin.py`
 - Code:

```
1 # Write your python code here
2 print("Learn security with TryHackMe!")
3
4 print 21 + 43
5
6
7
8
```
 - Run Code button
 - Exercise Complete! The flag is: `THM{ADDITION}`
 - Python code output:

```
Learn security with TryHackMe!
64
```

Bottom Screenshot (Room progress: 16%):

- Left Panel (Questions):**

Not Equal to	<code>!=</code>
Greater than or equal to	<code>>=</code>
Less than or equal	<code><=</code>

Answer the questions below

In the code editor, print the result of 21 + 43. What is the flag?

THM{ADDITION}

Print the result of 142 - 52. What is the flag?
- Right Panel (Code Editor):**
 - Files: `script.py`, `flag.txt`, `shipping.py`, `bitcoin.py`
 - Code: (Same as top screenshot)
 - Run Code button
 - Exercise Complete! The flag is: `THM{ADDITION}`
 - Python code output: (Same as top screenshot)
 - Notification: `Woop woop! Your answer is correct`

I typed `print(142-52)` in the code editor and got the flag.

The screenshot shows the TryHackMe Python Basics room interface. On the left, the 'Room progress (22%)' section lists three exercises. The first exercise, 'Print the result of 142 - 52. What is the flag?', is marked as 'Correct Answer' with a green checkmark. The second exercise, 'Print the result of 10 * 342. What is the flag?', is currently in progress with an empty input field and a 'Submit' button. The third exercise, 'Print the result of 5 squared. What is the flag?', is also in progress. On the right, the 'Python Code Editor' shows a script named 'script.py' with the following code:

```
1 # Write your code here
2 print("Learn security with TryHackMe!")
3
4 print 142-52
5
6
7
8
```

The output of the code is displayed below the editor:

```
Python code output
Learn security with TryHackMe!
90
```

A green notification bubble at the top right of the code editor says: 'Woop woop! Your answer is correct'.

I typed `print(10*342)` in the code editor and got the flag.

The screenshot shows the TryHackMe Python Basics room interface after the second exercise is completed. The 'Room progress (27%)' section now shows the first two exercises as 'Correct Answer'. The second exercise, 'Print the result of 10 * 342. What is the flag?', is now marked as 'Correct Answer' with a green checkmark. The third exercise, 'Print the result of 5 squared. What is the flag?', is still in progress. On the right, the 'Python Code Editor' shows the same script as before, but the code has been updated to:

```
1 # Write your code here
2 print("Learn security with TryHackMe!")
3
4 print 10*342
5
6
7
8
```

The output of the code is displayed below the editor:

```
Python code output
Learn security with TryHackMe!
3420
```

A green notification bubble at the top right of the code editor says: 'Woop woop! Your answer is correct'.

I typed `print(5**2)` in the code editor and got the flag.

The screenshot shows the TryHackMe Python Basics room interface. On the left, three exercises are listed, each with a 'Correct Answer' button and a 'Hint' button. The first exercise asks for the result of 142 - 52, with the answer `THM{SUBTRACT}`. The second asks for the result of 10 * 342, with the answer `THM{MULTIPLICATION_PYTHON}`. The third asks for the result of 5 squared, with the answer `THM{EXPONENT_POWER}`. On the right, a code editor shows a Python script that prints a message and calculates 5 squared. A notification bubble says 'Woop woop! Your answer is correct'. Below the code editor, the output shows 'Learn security with TryHackMe!' and '25'. A task list at the bottom shows 'Task 5 Logical and Boolean Operators' and 'Task 6 Shipping Project Introduction to If Statements'.

Room progress (33%)

✓ Correct Answer Hint

Print the result of 142 - 52. What is the flag?

THM{SUBTRACT}

✓ Correct Answer

Print the result of 10 * 342. What is the flag?

THM{MULTIPLICATION_PYTHON}

✓ Correct Answer

Print the result of 5 squared. What is the flag?

THM{EXPONENT_POWER}

✓ Correct Answer Hint

script.py

Woop woop! Your answer is correct

```
1 # Write y
2 print("Learn security with TryHackMe!")
3
4 print 5**2
5
6
7
8
```

Exercise Complete! The flag is: THM{EXPONENT_POWER}

Python code output

Learn security with TryHackMe!
25

THM AttackBox Python Coder

On the code editor, I created a new variable called Height and set its initial value to 200. I then added 50 to the "Height" variable and printed the new value of "Height" and got the flag.

The screenshot shows the TryHackMe Python Basics room interface. On the left, two exercises are listed, each with a 'Correct Answer' button. The first exercise asks to add 50 to the height variable, with the answer 'No answer needed'. The second asks to print the value of height, with the answer `THM{VARIABLES}`. On the right, a code editor shows a Python script that sets a height variable to 200, adds 50 to it, and prints the result. A notification bubble says 'Woop woop! Your answer is correct'. Below the code editor, the output shows 'Learn security with TryHackMe!' and '250'. A task list at the bottom shows 'Task 5 Logical and Boolean Operators' and 'Task 6 Shipping Project Introduction to If Statements'.

Room progress (50%)

On a new line, add 50 to the height variable.

No answer needed

✓ Correct Answer

On another new line, print out the value of height. What is the flag that appears?

THM{VARIABLES}

✓ Correct Answer

Task 5 Logical and Boolean Operators

Task 6 Shipping Project Introduction to If Statements

script.py

Woop woop! Your answer is correct

```
1 # Write y
2 print("Learn security with TryHackMe!")
3 height = 200
4 height = height + 50
5 print (height)
6
7
8
```

Exercise Complete! The flag is: THM{VARIABLES}

Python code output

Learn security with TryHackMe!
250

THM AttackBox Python Coder

In this task, we need to write a short program to calculate the total cost of a customer's shopping basket, including the cost of shipping

The screenshot shows the TryHackMe Python Basics room interface. On the left, a challenge titled 'No answer needed' is displayed. It contains a 'Complete' button and a text box with the placeholder 'THM{IF_STATEMENT_SHIPPING}'. Below this, a green 'Correct Answer' button and an orange 'Hint' button are visible. The hint text reads: 'In shipping.py, on line 15 (when using the Code Editor's Hint), change the **customer_basket_cost** variable to **101** and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.' A 'Submit' button is at the bottom of the hint section. On the right, the 'Python Coder' tab is active, showing a Python script in 'shipping.py'. The script calculates the total cost based on a customer's basket cost and weight. A green notification bubble says 'Woop woop! Your answer is correct'. Below the code, the output shows '86.8'. A green banner at the bottom of the code editor states 'Exercise Complete! The flag is: THM{IF_STATEMENT_SHIPPING}'.

```
script.py
13 customer_
14
15 # Write if statement here to calculate the total cost
16 shipping_cost=0
17 if customer_basket_cost < 100:
18     shipping_cost=1.2 * customer_basket_weight
19 print (customer_basket_cost+shipping_cost)
20
```

Exercise Complete! The flag is: THM{IF_STATEMENT_SHIPPING}

Python code output

86.8

The next step is for me to change the value in line 15, as shown in the image below, to 101 and then run the code again.

The screenshot shows the TryHackMe Python Basics room interface after the second step. The challenge on the left now shows the 'Correct Answer' button. The hint text remains the same. The 'Python Coder' tab on the right shows the same Python script, but the value '100' in line 17 has been changed to '101'. A green notification bubble says 'Woop woop! Your answer is correct'. Below the code, the output shows '101'. A green banner at the bottom of the code editor states 'Exercise Complete! The flag is: THM{MY_FIRST_APP}'.

```
script.py
13 # write if statement here to calculate the total cost
16 shipping_cost=0
17 if customer_basket_cost < 101:
18     shipping_cost=1.2 * customer_basket_weight
19 print (customer_basket_cost+shipping_cost)
20
21
22
```

Exercise Complete! The flag is: THM{MY_FIRST_APP}

Python code output

101

I learnt about for and while loops. The task here is to write a program that prints out the numbers from 0-50. I wrote the code as shown in the editor and got the flag. In order to do that, I specified the range by inputting 51, as in programming, we start indexing from 0.

The screenshot shows the TryHackMe Python Basics room interface. On the left, a text box explains the range function and provides a code example:

```
for i in range(5):  
    print(i)
```

. Below this, a task instruction says: "On the code editor, click back on the 'script.py' tab and code a loop that outputs every number from 0 to 50." A text input field contains the flag `THM{LOOPS_WHILE_FOR}`, with "Correct Answer" and "Hint" buttons. On the right, the code editor shows a Python script:

```
1 # Write your python code here  
2 print("Learn security with TryHackMe!")  
3 for i in range (51):  
4     print (i)  
5  
6  
7  
8
```

. The output shows the numbers 0 through 6. A green banner at the top of the editor says "Exercise Complete! The flag is: THM{LOOPS_WHILE_FOR}".

For this task I opened the bitcoin.py file in the code editor. I then defined a function called bitcoinToUSD with the parameters bitcoin_amount and bitcoin_value_usd. Inside the function, the program calculates the USD value by multiplying bitcoin_amount by bitcoin_value_usd and returns the calculated USD value. The program also checks if the returned value is below \$30,000 and prints an alert message if the value is below \$30,000.

The screenshot shows the TryHackMe Python Basics room interface for a second task. On the left, a text box explains the task: "value of Bitcoin falls below a particular value in dollars. In the code editor, click on the bitcoin.py tab. Write a function called **bitcoinToUSD** with two parameters: **bitcoin_amount**, the amount of Bitcoin you own, and **bitcoin_value_usd**, the value of bitcoin in USD. The function should return **usd_value**, which is your bitcoin value in USD (to calculate this, in the function, you times bitcoin_amount variable by bitcoin_value_usd variable and return the value). The start of the function should look like this:" followed by a code snippet:

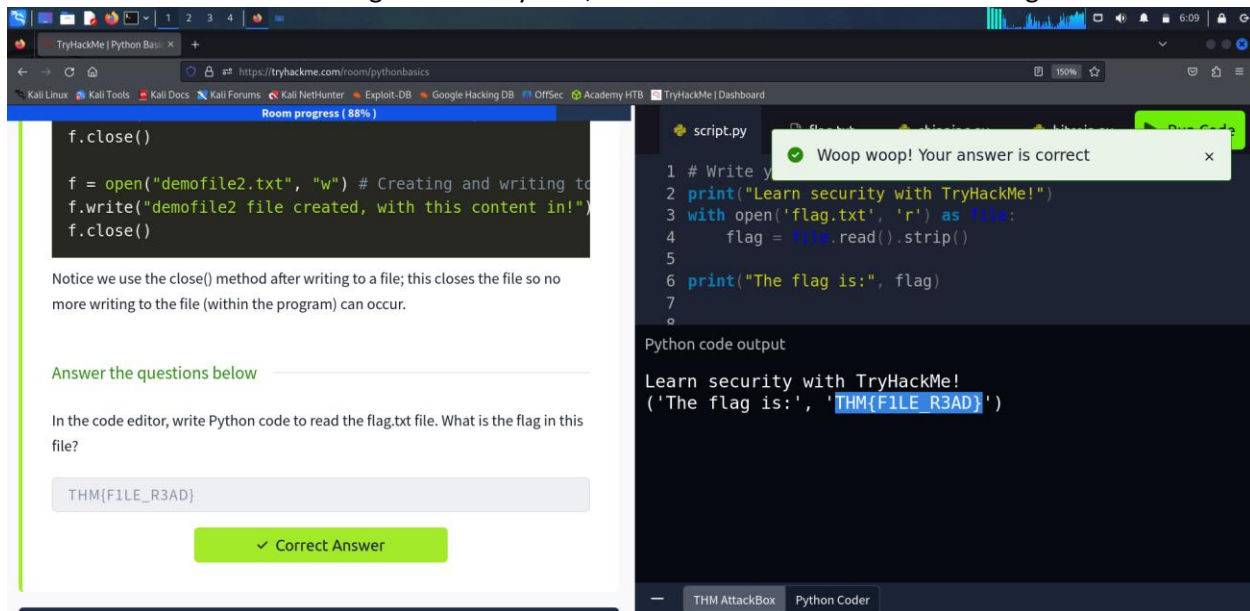
```
def bitcoinToUSD(bitcoin_amount, bitcoin_value_usd):
```

. Below this, instructions say: "Once you've written the bitcoinToUSD function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below \$30,000; if it does, output a message to alert you (via a print statement)." A text input field contains the flag `THM{BITCOIN_INVESTOR}`, with "Correct Answer" and "Hint" buttons. On the right, the code editor shows the following Python code:

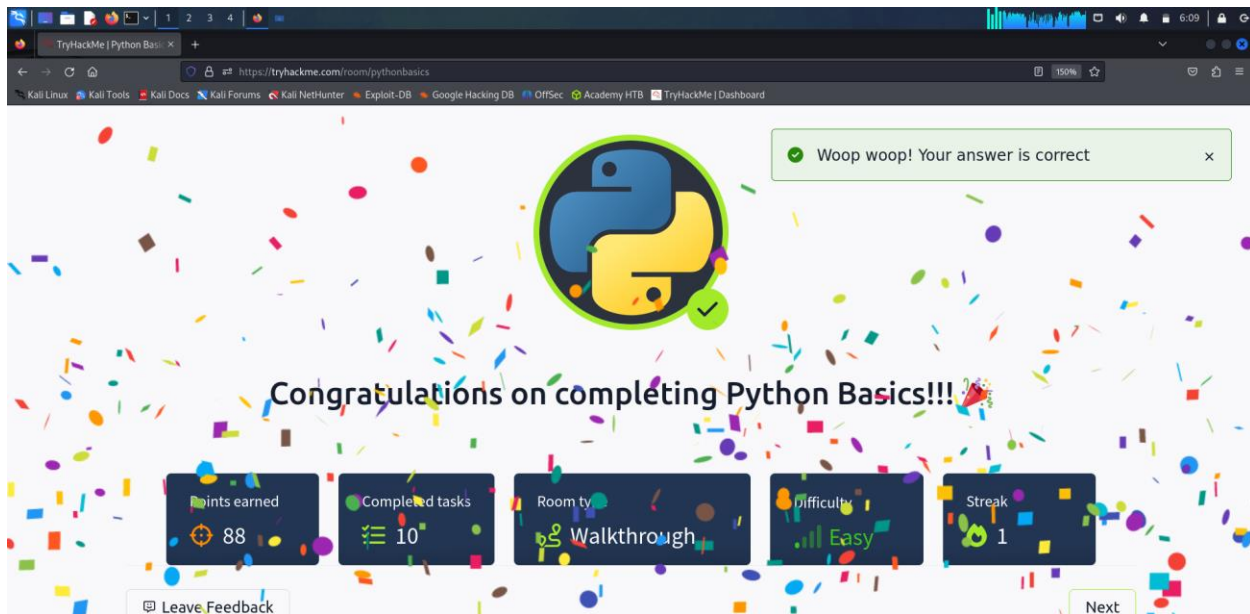
```
19 return bitcoin_amount * bitcoin_value_usd  
20  
21 # 2) use function to calculate if the investment is below  
22 $30,000  
23 my_investment =  
24 bitcoinToUSD(investment_in_bitcoin,bitcoin_to_usd)  
25 if my_investment < 30000:  
26     print ("Investment is less than 30000")
```

. A green banner at the top of the editor says "Exercise Complete! The flag is: THM{BITCOIN_INVESTOR}".

To read the contents of the flag.txt file in Python, I used the code as shown in the image below



3. Shareable Link and Final Screenshot



4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fpythonbasics%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

https://www.facebook.com/sharer/sharer.php?u=https%3A%2F%2Ftryhackme.com%2Froom%2Fpythonbasics%3Futm_source%3Dfacebook%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&t=Share%20your%20achievement

https://www.linkedin.com/sharing/share-offsite/?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fpythonbasics%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

In conclusion, learning Python programming has provided me with solid foundation in fundamental concepts crucial for both programming and cybersecurity. Tasks covering variables, loops, functions, data structures, if statements, and file handling have demonstrated the versatility and power of Python. From executing simple tasks like printing "Hello World" to performing complex operations and managing variables, Python's capabilities are evident. I have learnt Python's relevance in cybersecurity, such as offering the ability to script and automate tasks, analyze data, and conduct security testing. In summary, the journey of learning Python programming has equipped me with the knowledge, skills, and confidence necessary to navigate the cybersecurity landscape.